

Prediction of Bank Term Deposit Subscription Using Data Mining Classification Algorithms

Abdalkarim M. S. Alaraj
Dept. of Artificial Intelligence and Data
Engineering
International University of Sarajevo
aalaraj@student.ius.edu.ba

Muhamed Maglić
Dept. of Artificial Intelligence and Data
Engineering
International University of Sarajevo
mmaglic@student.ius.edu.ba

Hamza Hubanić
Dept. of Artificial Intelligence and Data
Engineering
International University of Sarajevo
hhubanic@student.ius.edu.ba

Faculty of Engineering and Natural Sciences (FENS) — Supervisor: Prof. Emine Yaman (eyaman@ius.edu.ba)

In today's world, banks are still using personalized marketing, usually phone calls, to reach their potential clients and pitch financial products, in this case term deposit subscriptions. Since finding customers that will subscribe requires brute force calling of multiple audiences, it is a very inefficient, expensive and slow approach. Therefore, predicting potential customers ahead of time, creating market segmentation and finding loyal clients would be a better solution, which is the goal of our project. To reach those objectives, we needed a real-world bank dataset with a substantial number of records and attributes for meaningful analysis. Thus, we used the Bank Marketing dataset collected from a Portuguese bank with 45,000 records and 16 attributes. The target attribute, telling if a customer is subscribed for a term deposit, is binary (yes/no), which was a perfect fit for our binary classification models. We tested and compared five models that we covered in our Data Mining class: Decision Tree, Naive Bayes, k-Nearest Neighbours, Random Forest, and Logistic Regression. Following machine learning pipeline practices, we started with data preprocessing methods like target encoding, removing duplicates and redundant attributes, addressing missing values, scaling numerical features and encoding categorical attributes. Afterwards, exploratory data analysis was conducted to reveal data patterns and meaningful connections, and finally we moved to training and testing our classification models. All models achieved around 89% accuracy, but since the dataset is heavily imbalanced, F1-score and recall were the most relevant metrics for finding actual subscribers. ROC-AUC was also important for ranking the most promising clients. All models were able to contribute to meaningful discoveries, which we will see in the following sections.

Keywords— data mining; classification; bank marketing; term deposit; imbalanced data; Random Forest; Naive Bayes; ROC-AUC.

I. INTRODUCTION

Marketing is one of the main ways that banks attract new customers and sell their products. A very common product is the term deposit, where a client agrees to keep a fixed amount of money in the bank for a certain period in exchange for an interest rate. To sell these deposits, banks run direct marketing campaigns, and one of the most used methods is calling clients on the phone [1]. However, calling clients is expensive and takes a lot of time, and most of the clients that are contacted do not end up subscribing. Because of this, it would be very useful for a bank to know in advance which clients are more likely to say yes, so that the marketing team can focus its effort on them.

This kind of problem is a good fit for data mining. Data mining is process of finding useful patterns and knowledge from large volumes of data [2]. One of the main tasks in data mining is classification, where the goal is to assign each record to one of predefined classes [3]. In our case, two classes are simple: client either subscribes to term deposit (yes) or does not (no). Classification is supervised task, meaning model learns from labeled examples where correct class is already known, and then predicts class of new unseen cases. This fits bank context well, since banks already have records from previous campaigns showing who subscribed and who did not. In this project we use Bank Marketing dataset, collected from Portuguese bank and publicly available through UCI Machine Learning Repository [4]. Dataset contains client details, information about current campaign and summary of previous ones. Our objective is to construct and compare a few classification models that estimate whether a client will subscribe to a term deposit, as

well as to figure out which client's traits matter the most. We picked five classification approaches that were covered during our Introduction to Data Mining course: Decision Tree, Naive Bayes, k-Nearest Neighbors (k-NN), Random Forest, and Logistic Regression. We chose these five because they come from different families of methods (tree-based, probabilistic, distance-based, ensemble, and linear), so comparing them is more interesting than comparing very similar models. In addition to classification, we also applied K-Means clustering to group similar clients, and we used the Random Forest model to find the most important features.

The rest of this paper is organized as follows. Section II reviews related work. Section III describes the dataset, the preprocessing steps, the five algorithms, and the evaluation metrics. Section IV presents the results and discusses them, including the exploratory analysis, the model comparison, the feature importance, and the clustering. Section V gives the conclusion, and Section VI lists possible future work.

II. RELATED WORK

The Bank Marketing dataset has been studied by several researchers. The dataset itself was introduced by Moro, Cortez, and Rita [4], who used a data-driven approach to predict the success of bank telemarketing. In an earlier work, Moro, Laureano, and Cortez [5] applied the CRISP-DM methodology to the same kind of bank marketing data and showed that data mining can support marketing decisions. Elsalamony [6] analyzed the same dataset using several data mining techniques, including neural networks and decision trees, and reported that handling the imbalance of the classes is an important issue for this problem.

Comparing several classification algorithms on a single dataset is a common thing in both student's and research projects. To illustrate, in the medical domain, several studies have compared Naive Bayes, Logistic Regression, Random Forest, and other related approaches to forecast heart disease [7], and they usually find out that no single algorithm is always the best. Although this approach of training and testing different algorithms is useful, most of the works we examined on this topic are still emphasizing that dataset quality and data processing quality are still foundation of creating reliable models

One of major problems in datasets like ours is class imbalance. In direct marketing, only a fraction of customers will actually subscribe and take an offer. Consequently, dataset will have more class "no" records and models will be biased towards majority class. He and Garcia [22] refers that standard classifiers, like ones we used, trained on imbalanced data will always lean to majority class, resulting in high accuracy, but weak precision, which can be deceptive. To alleviate this biasness, different techniques were proposed. Most popular is SMOTE, the oversampling method introduced by Chawla et al. [24], this method generates new synthetic samples for the minority class.

Motivated by approach of comparing classifiers from conducted research on similar problems, our work follows the same path, concentrating on bank marketing problem while being aware of that dataset is imbalanced so that we can create solution around these constraints. Rather than relying only on accuracy we gave more attention to precision, recall, F1-score and AUC-ROC. Following these metrics we ordered models based on the needs and objectives of bank marketing. Additionally, we added clustering and feature-importance analysis to provide value for understanding characteristics of customers and their associations.

III. MATERIALS AND METHODS

A. Dataset Description

The Bank Marketing dataset has 45,000 records, 16 decision attributes and one target attribute. Each record represent one previous customer with all collected data through out interaction with bank. Customer attributed naturally divides into three categories details about the client, information about last interaction with the current campaign, and data collected from previous campaigns. Table I shows all the attributes and what they mean.

TABLE I. Description of the Dataset Attributes

Attribute	Type	Description
age	numeric	Client age in years
job	categorical	Type of job (admin, technician, retired, ...)
marital	categorical	Marital status
education	categorical	Education level
default	categorical	Has credit in default?

balance	numeric	Average yearly balance (euros)
housing	categorical	Has a housing loan?
loan	categorical	Has a personal loan?
contact	categorical	Contact communication type
day	numeric	Last contact day of the month
month	categorical	Last contact month
duration	numeric	Last contact duration (seconds)
campaign	numeric	Number of contacts in this campaign
pdays	numeric	Days since last contact (-1 = never)
previous	numeric	Number of contacts before this campaign
poutcome	categorical	Outcome of the previous campaign
y (target)	categorical	Subscribed to term deposit? (yes / no)

Attribute y is the target class attribute we want to predict. As can be seen in Fig. 1, the dataset is heavily imbalanced with only 11.7 % of customers subscribed to term deposit while remaining majority is not-subscribed. This class imbalance required from us different technical approach when it comes to data processing and model evaluation.

B. Data Preprocessing

Real datasets representing scenarios like our are always coming with its inconsistencies and requires preprocessing treatment. Thus, preprocessing is an essential step before training any model [2]. We applied the following steps in order.

1. Target encoding: We mapped the target values yes to 1 and no to 0, classification models work mathematically, we need numbers instead of strings.
2. Removing the duration attribute: The duration attribute represents time of last phone call occurred in seconds. Since this information is irrelevant for models predicting future customers, we removed it.
3. Handling the pdays negative value: In the pdays attribute, the value -1 does not actually mean a real count of days. It means the client was never contacted in a previous campaign. So, we replaced -1 with 0, and we also created a new binary variable, `was_contacted_before`, which can be valuable information.
4. Missing values: A few attributes (like job, education, contact, and poutcome) include missing entries. The missing data is loaded as NaN values, mainly related to customers who were not part of previous campaigns. We did not delete this rows, since one-hot encoding applied to these variables transforms categories into dummy columns with 0 and 1 occurrence, therefore missing values retained their meaning.
5. Duplicate removal: We got rid of the duplicate records, preventing models to detect these rows as more important than others.
- 6) Outliers: We inspected the numeric attributes such as balance and campaign with boxplots. Many of the extreme

values are real clients (for example, a client with a very high balance), not errors, so we decided not to remove them. The scaling step below reduces their effect.

7) One-hot encoding: The categorical attributes were converted into numeric form using one-hot encoding, because algorithms such as Logistic Regression, k-NN, and Naive Bayes need numeric input. After encoding, the number of features increased.

8) Train/test split: We split the data into 75% for training and 25% for testing. We used stratified sampling so that both parts keep the same proportion of subscribers as the full dataset.

9) Feature scaling: We standardized the numeric features so that they all have a similar range. This is important for k-NN and Logistic Regression, which are sensitive to the scale of the values. Decision Tree and Random Forest do not need scaling, but it does not harm them.

C. Classification Algorithms

We used five classification algorithms. Each one comes from a different family, which makes the comparison more meaningful.

1) Decision Tree: A decision tree splits the data step by step using simple rules on the attributes, building a tree where each leaf gives a class [8], [9]. Trees are easy to understand and explain, which is useful for non-technical staff. Their main weakness is that if they grow too deep, they can overfit the training data [3]. We decided to limit the maximum depth of the tree, a bit more than usual, in order to have kind of control over this.

2) Naive Bayes: A probabilistic classifier grounded in Bayes' theorem [10]. It basically takes the attributes as if they are independent of one another given the class label, which is often not actually the case, but somehow it still behaves well in real usage, and it is also fast [11]. We applied the Gaussian form, since it fits numeric attributes nicely.

3) k-Nearest Neighbors (k-NN): Classifies a new client by looking at the k most similar clients (its nearest neighbors) and taking a majority vote [12]. It does not build a model during training; instead, it stores the data and computes distances when predicting. Because it relies on distances, scaling the features is very important [3]. We used $k = 15$.

Random Forest: An ensemble technique that builds many decision trees from different parts of data giving prediction of target attribute based on most voted class [13], [14]. Observing average of these trees is more reliable than putting all trust into one decision tree. Also, more trees can detect non-linear patterns while avoiding possibility of overfitting. Moreover, it can also reveal the most important attributes. The downside is longer running time and more complex interpretability.

Logistic Regression: Logistic regression is the simple linear model for binary classification [15]. It shows probability that client belongs to positive class, in this case as term deposit subscriber. It is simple to build, run and interpret, showing coefficients of each attribute to reveal *their*

importance. On the other hand, because of simplicity, the model is struggling to detect non-linear and less obvious patterns occurring in real-world data.

D. Clustering Method

In addition to classification, we used K-Means clustering [16], [17] to explore the data without using the target. K-Means groups the clients into k clusters so that clients in the same cluster are similar to each other. We used the elbow method to choose the number of clusters and then projected the data into two dimensions using Principal Component Analysis (PCA) [18] so that the clusters could be visualized.

E. Evaluation Metrics

Because the dataset is imbalanced, accuracy alone is not enough to judge the models. A model that always predicts “no” would already reach about 88% accuracy without learning anything useful. For this reason, we also used precision, recall, F1-score, and ROC-AUC [19], [20]. These metrics are computed from the confusion matrix, which counts the true positives, false positives, true negatives, and false negatives. Precision measures how many of the clients predicted as subscribers really subscribed. Recall, also called the true positive rate, measures how many of the real subscribers the model managed to find. There is usually some tradeoff between the two making a model predict “yes” more often increases recall, but it also lowers precision. The F1-score is the harmonic mean of precision and recall, and it gives one balanced number [21]. ROC-AUC measures how well the model separates the two classes across all possible thresholds, and it is known to be useful for imbalanced data [22], [23]. We also recorded the training time of each model, since training speed can matter when a model has to be retrained often.

IV. RESULTS AND DISCUSSION

A. Exploratory Data Analysis

Before training the models, we explored the data with several plots. Fig. 1 shows the distribution of the target. It is clearly imbalanced: the number of clients who did not subscribe is much larger than the number who did.

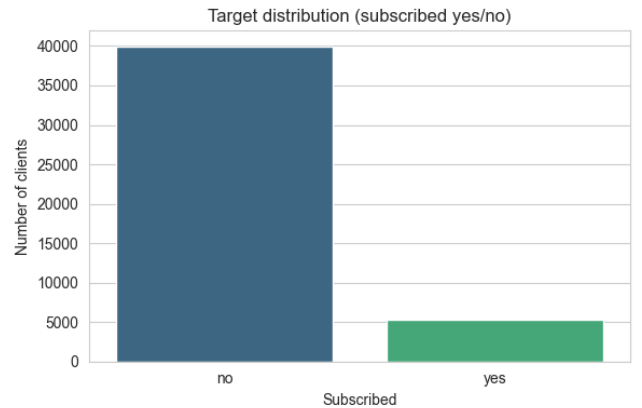


Fig. 1. Distribution of the target variable (subscribed yes/no).

Fig. 2 shows the correlation between the numeric features. Most correlations are weak, which means the features carry

different information and there is little redundancy. The strongest correlations are between pdays, previous, and the engineered was_contacted_before feature, which all describe the history of previous contacts. We can also see that no single numeric feature has a strong linear correlation with the target y (the highest is only about 0.17), which already suggests that the prediction will not be easy.

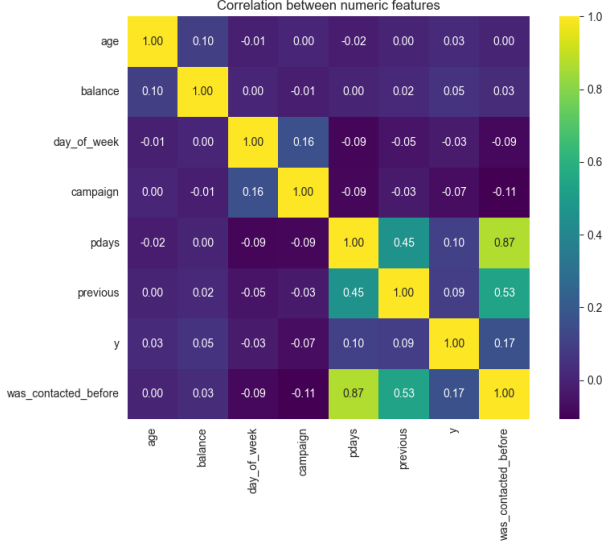


Fig. 2. Correlation heatmap of the numeric features.

Fig. 3 shows the subscription rate by the outcome of the previous campaign (poutcome). Clients whose previous campaign was a success subscribed again at a very high rate, about 64%, which is far higher than for clients whose previous outcome was “failure” or “other” (around 12–16%). This makes sense: a client who accepted an offer before is more likely to accept again.

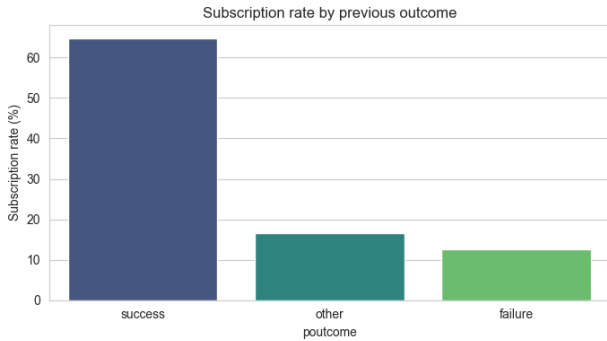


Fig. 3. Subscription rate by previous campaign outcome.

We also observed the distributions of the numeric features, kind of broad. The age of the clients is pretty much bell-shaped, and it sits around the working-age range, but balance, campaign, and previous are strongly right-skewed, like noticeably. Most clients end up with a modest balance and only a few contacts, yet there are a handful of cases with extremely large values. This skew is basically why we standardized the numeric features. And it’s also why we didn’t delete extreme values, because they are still real clients. When we compared subscribers versus non-subscribers, the subscribers usually had a slightly higher

balance. Meanwhile, clients who were contacted many times in the current campaign were less likely to subscribe, so repeatedly calling a reluctant person has little or no benefit.

B. Classification Results

Table II shows the performance of the five models on the test set. All models reach a similar accuracy of about 89%, except Naive Bayes, which is a bit lower at 85.6%. However, as explained before, accuracy is misleading on this imbalanced dataset, so we focus on the other metrics.

TABLE II. Performance of the Five Classifiers

Model	Acc	Prec	Rec	F1	AUC
Naive Bayes	0.856	0.394	0.424	0.409	0.736
Decision Tree	0.893	0.608	0.228	0.331	0.655
Random Forest	0.894	0.627	0.225	0.331	0.771
k-NN	0.894	0.633	0.215	0.321	0.729
Logistic Reg.	0.894	0.670	0.181	0.285	0.754

What is interesting about results is that none of our models wins by all metrics. Naive Bayes has the lowest accuracy, but it also has the highest recall (0.424) and the highest F1-score (0.409). So basically, that suggests Naive Bayes is best at finding the clients who will subscribe, even if it also triggers more false alarms. The other four models shows lower performance on finding actual subscribers, with recall no higher then 0.228, meaning that only 22.8% percent of actual subscribers are discovered at most. Having high accuracy and precision comes from reliably predicting majority class which is not useful in this case. Logistic Regression has the highest precision (0.670) but the lowest recall (0.181). That means when it predicts “yes” it is likely correct, however it says “yes” far too rarely. Decision Tree, Random Forest, and k-NN are in the middle, with fairly similar F1-scores around 0.32–0.33.

Fig. 4 shows the ROC curves for all five models. Here it is a bit clearer: Random Forest has the highest ROC-AUC of 0.771, then Logistic Regression 0.754 and after that Naive Bayes 0.736. The Decision Tree has the lowest ROC-AUC (0.655). A higher ROC-AUC means that, if we used the model’s predicted probability to rank clients from most to least likely to subscribe, Random Forest would give the best ranking. This is very useful in practice, because a bank could simply call the top clients on the ranked list first.

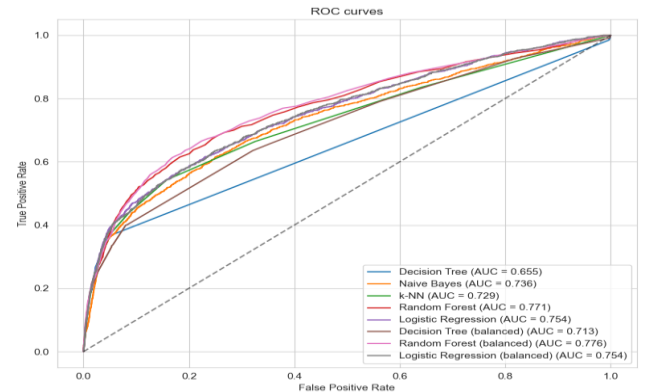


Fig. 4. ROC curves of all models, including the class-weighted versions.

Putting these results together, the “best” model depends on the goal. If the bank wants to catch as many potential subscribers as possible at the default threshold, Naive Bayes is the most useful because of its higher recall and F1. If the bank wants to rank clients and call the most promising ones first, Random Forest is the best because of its higher ROC-AUC. The low recall of most models also shows that the imbalance should be handled directly, which we discuss in the future work.

Regarding training time, k-NN was the fastest to train (0.01 s) because it does not build a model during training, while Random Forest was the slowest (1.19 s) because it builds many trees. However, speed is not a concern at this dataset size, since all training times are very small.

It is also worth comparing our results with the related work and whatever is similar. Studies that use the same dataset often report accuracy values in a comparable interval to ours, like 88–90% [6], so this sort of a confirmation that our pipeline behaves as intended. The recall being low shows what He and Garcia [22] already discussed about the imbalance issue. For that reason, relying solely on accuracy for evaluating model performance, like some earlier researches had done, can be deceptive and cover the real picture. Therefore, using mix of metrics, like we did, gives more insights and stronger assessments for our objective.

C. Summary of the Comparison

To make comparison final and easy to interpret, table III summarizes which model is most successful for given metric. Using this table we can reach our objectives to deliver data-driven decisions for bank marketing which was our final goal. If bank need to catch as many subscribers as possible naïve bayes is most effective algorithm. On the other hand, if bank has limited budget for marketing and do not want to lose money on calling customers who will not subscribe, Logistic regression is best performing for that case. And if the goal is to arrange clients for a calling list, the ROC-AUC column indicates Random Forest.

TABLE III. Best Model per Evaluation Metric

Metric	Best Model
Accuracy	Logistic Regression (0.894)
Precision	Logistic Regression (0.670)
Recall	Naive Bayes (0.424)
F1-score	Naive Bayes (0.409)
ROC-AUC	Random Forest (0.771)
Training speed	k-NN (0.01 s)

D. Feature Importance

Fig. 5 shows the 15 most important features from the Random Forest model. The top features are balance, age, and day, followed by campaign and the previous-outcome-success feature. This somewhat matches the exploratory analysis; the success of the previous campaign ranks among the top features, which aligns with Fig. 3. Meanwhile, the numeric features balance and age rank very high, indicating that the client's financial situation and age are significant

factors. The lower rankings of features like housing and education suggest that the client's personal and financial profile is more important than the specifics of a single call.

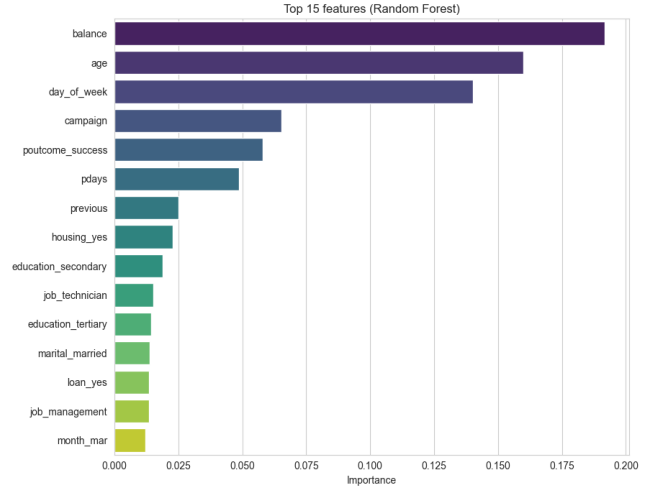


Fig. 5. Top 15 features by Random Forest importance.

E. Clustering Results

Finally, we applied K-Means clustering with four clusters on the numeric features and projected the result into two dimensions using PCA. Fig. 6 shows the clusters. Four distinct groups can be observed: "Older, high balance" clients spread on the right side, "Younger, low balance" forming the dense central mass representing majority of clients, "Over-contacted (low subscribe)" on the left side indicating clients contacted many times but rarely subscribed, and "Previously engaged (high subscribe)" representing clients with prior campaign contact who showed higher subscription rate. The clustering is exploratory and does not incorporate the target, but it supports the same conclusion from classification and feature-importance analysis: financial situation and contact history are the main factors that distinguish different types of clients.

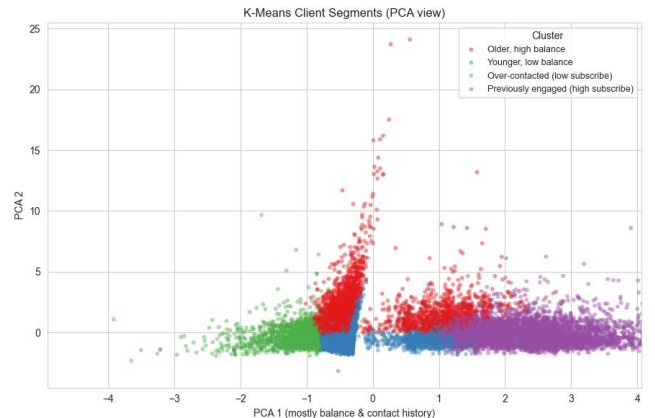


Fig. 6. K-Means clusters shown in the PCA projection.

From a practical point of view, these results have clear meaning for the bank. Because the models can sort clients by their predicted probability of subscribing, the bank does not have to treat the prediction as a strict yes/no decision. Instead, it can use the ranking to build a calling list, starting

with the clients who have the highest predicted probability. Even if the recall at the default threshold is low, a good ranking (high ROC-AUC) still helps the marketing team to reach a large share of the future subscribers while calling only part of all clients. This is basically the cost saving thing that motivated the project in the first place, and it also indicates that the usefulness of a model depends on how it is used, not only on one single accuracy number.

F. Effect of Class Weighting

The low recall of the original models tells us that they miss significant number of subscribers. To alleviate this problem, we included option `class_weight = "balanced"` on Decision Tree, Random Forest and Logistic Regression, since only they have that option out of five tested. The core logic behind it is to make minority class count matters more during training. Table IV compares the original against the weighted versions.

TABLE IV. Original vs. Class-Weighted Models

Model	Acc	Prec	Rec	F1	AUC
Decision Tree	0.893	0.608	0.228	0.331	0.655
Decision Tree (bal.)	0.856	0.389	0.402	0.395	0.713
Random Forest	0.894	0.627	0.225	0.331	0.771
Random Forest (bal.)	0.893	0.644	0.198	0.303	0.776
Logistic Reg.	0.894	0.670	0.181	0.285	0.754
Logistic Reg. (bal.)	0.757	0.264	0.603	0.368	0.754

Effect of class weighting is fairly noticeable. For Logistic Regression, recall jumps from 0.181 to 0.603, so model now finds around 60% of real subscribers instead of only 18%, but cost is that precision falls from 0.670 to 0.264 and accuracy drops to 0.757. This happens because model predicts "yes" more often, even if that means more false alarms. Decision Tree also improves moderately. Recall goes from 0.228 to 0.402, F1 rises from 0.331 to 0.395, and ROC-AUC moves from 0.655 to 0.713. Random Forest does not change much, and its F1 even drops slightly, which suggests that class weighting is not universal solution and depends on specific model and dataset.

Although applying weighting to other models, Naive Bayes still comes out with best F1 (0.409) among all eight, confirming it is naturally good fit for imbalanced situation. Moreover ROC-AUC numbers does not change much with weighting, which is expected since weighting mostly affects behavior around decision threshold, not model's ability to rank clients. Overall, class weighting is straightforward and effective approach when goal is to detect more subscribers, assuming bank can tolerate lower precision that comes with it.

G. Limitations

Our analysis has few limitations. Firstly, we used default threshold of 0.5 for all models without tuning it, this cause relatively low recall on all models, and can be modified to marketing needs. Moreover, we only applied class

weighting for data imbalance, we did not cover other imbalance strategies like oversampling. Furthermore, we relied mostly on default hyperparameters and only adjusted things like tree depth and value of k , so models could probably perform better with proper tuning. Finally, dataset comes from one bank in one country, so results might not carry over well to other banks or different regions.

V. CONCLUSION

Working through out this project we tested five different classification algorithms on the bank marketing dataset, also testing class weighting afterwards to handle imbalance and compare to prior results. We applied a complete preprocessing pipeline and compared the models using several metrics. What is mainly noticeable on an imbalanced dataset, a high accuracy can be misleading: although four of the models reached about 89% accuracy, most of them missed the majority of the real subscribers. When we looked at recall and F1-score, Naive Bayes had the best findings, while Random Forest achieved the best ROC-AUC and would be the best choice for ranking clients. Adding the class weight attribute to the Decision Tree and Logistic Regression increased their recall strongly, with Logistic Regression going from finding 18% to 60% of the subscribers, at the cost of lower precision. The feature-importance and clustering analyses both similarly pointed to the client's balance, age, and contact history as the most important factors. Overall, this project showed how important preprocessing, the handling of class imbalance, and the right choice of evaluation metric are in a real data mining analysis.

VI. FUTURE WORK

There are several ways this could be extended. In this project we have tested class weighting. The natural next step is to compare it with oversampling methods like SMOTE [24] to see which approach handles the imbalance best. The decision threshold could also be tuned so that the bank can choose for itself the balance between precision and recall depending on the cost of a missed client. The models can be improved in future with proper hyperparameter tuning. Finally, other algorithms such as SVM or boosting methods could be added to the comparison.

REFERENCES

- [1] S. Moro, P. Cortez, and P. Rita, "A data-driven approach to predict the success of bank telemarketing," *Decision Support Systems*, vol. 62, pp. 22–31, 2014.
- [2] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Waltham, MA: Morgan Kaufmann, 2011.
- [3] P.-N. Tan, M. Steinbach, A. Karpatne, and V. Kumar, *Introduction to Data Mining*, 2nd ed. New York: Pearson, 2019.
- [4] D. Dua and C. Graff, *UCI Machine Learning Repository*. Irvine, CA: University of California, School of Information and Computer Science, 2019. [Online]. Available: <https://archive.ics.uci.edu/ml>
- [5] S. Moro, R. Laureano, and P. Cortez, "Using data mining for bank direct marketing: An application of the CRISP-DM methodology," in *Proc. European Simulation and Modelling Conf. (ESM)*, 2011, pp. 117–121.

- [6] H. A. Elsalamony, "Bank direct marketing analysis of data mining techniques," *International Journal of Computer Applications*, vol. 85, no. 7, pp. 12–22, 2014.
- [7] C. B. Latha and S. C. Jeeva, "Improving the accuracy of prediction of heart disease risk based on ensemble classification techniques," *Informatics in Medicine Unlocked*, vol. 16, 100203, 2019.
- [8] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, *Classification and Regression Trees*. Belmont, CA: Wadsworth, 1984.
- [9] J. R. Quinlan, "Induction of decision trees," *Machine Learning*, vol. 1, no. 1, pp. 81–106, 1986.
- [10] T. Bayes, "An essay towards solving a problem in the doctrine of chances," *Philosophical Transactions of the Royal Society of London*, vol. 53, pp. 370–418, 1763.
- [11] P. Domingos and M. Pazzani, "On the optimality of the simple Bayesian classifier under zero-one loss," *Machine Learning*, vol. 29, no. 2–3, pp. 103–130, 1997.
- [12] T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Transactions on Information Theory*, vol. 13, no. 1, pp. 21–27, 1967.
- [13] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [14] T. K. Ho, "Random decision forests," in *Proc. 3rd Int. Conf. Document Analysis and Recognition (ICDAR)*, 1995, pp. 278–282.
- [15] D. R. Cox, "The regression analysis of binary sequences," *Journal of the Royal Statistical Society: Series B*, vol. 20, no. 2, pp. 215–242, 1958.
- [16] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Mathematical Statistics and Probability*, vol. 1, 1967, pp. 281–297.
- [17] S. Lloyd, "Least squares quantization in PCM," *IEEE Transactions on Information Theory*, vol. 28, no. 2, pp. 129–137, 1982.
- [18] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York: Springer, 2002.
- [19] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.
- [20] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proc. 23rd Int. Conf. Machine Learning (ICML)*, 2006, pp. 233–240.
- [21] D. M. W. Powers, "Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation," *Journal of Machine Learning Technologies*, vol. 2, no. 1, pp. 37–63, 2011.
- [22] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.
- [23] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, e0118432, 2015.
- [24] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [25] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [26] C. R. Harris et al., "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.
- [27] W. McKinney, "Data structures for statistical computing in Python," in *Proc. 9th Python in Science Conf. (SciPy)*, 2010, pp. 56–61.
- [28] J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.
- [29] M. L. Waskom, "Seaborn: Statistical data visualization," *Journal of Open Source Software*, vol. 6, no. 60, 3021, 2021.
- [30] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York: Springer, 2009.
- [31] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York: Springer, 2006.
- [32] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning*. New York: Springer, 2013.
- [33] C. C. Aggarwal, *Data Mining: The Textbook*. Cham, Switzerland: Springer, 2015.
- [34] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.